Cross-Environment Behavioral Fingerprinting for IoT Device Security: A Cautionary Tale in Evaluation Methodology

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Abstract

This project set out to test whether behavioral network fingerprints for IoT devices transfer across different network environments, using two independent public datasets, IoT-23 and CICIoT2023, and three anomaly detection approaches: a per-feature statistical z-score baseline, a K-Means clustering detector, and an Isolation Forest. Our initial evaluation appeared to show that a simple statistical baseline achieves strong cross-environment transfer (F1 = 0.974) while more expressive models fail. Closer investigation revealed this apparent success was substantially an artifact of our evaluation protocol: standardizing test data against its own benign reference causes a simple threshold-based model's learned parameters to collapse to a fixed, training-data-independent point, making its result a measure of the test environment's own internal separability rather than genuine transfer. After correcting this, we found that none of the three models, under any tested feature subset, achieve genuine cross-environment transfer between these two datasets. We traced the root cause to incompatible flow-construction methodologies: CICIoT2023's packet-count feature is a fixed aggregation window artifact of its own published extraction pipeline, not a measurement of actual flow length, making it fundamentally incomparable to IoT-23's native connection-level statistics. A follow-up diagnostic further showed that this evaluation artifact itself affects models unevenly: Isolation Forest's cross-environment failure is a genuine, verified structural mismatch, while K-Means clustering is substantially (though not fully) subject to the same self-calibration degeneracy that fully explained the statistical baseline's apparent success. A complementary within-dataset experiment found that IoT-23's five capture scenarios are genuinely distinguishable by behavioral features alone (macro F1 = 0.88), driven overwhelmingly by connection duration rather than protocol type. We report this corrected, root-caused investigation, including the debugging process itself, as a methodological contribution: apparent cross-environment generalization in IoT anomaly detection evaluation should not be trusted without verifying that it is not an artifact of the evaluation protocol itself.

Keywords: *IoT security, behavioral fingerprinting, anomaly detection, cross-environment generalization, dataset shift, evaluation methodology, intrusion detection*

1. Introduction

IoT devices are, by design, repetitive on the network. A smart camera checks in with the same cloud endpoint on a predictable schedule, a thermostat sends small bursts of telemetry at fixed intervals, and a smart plug rarely contacts more than one or two destinations at all. This regularity is a security opportunity: once a device's normal behavior is characterized, deviations from it become a practical signal for detecting compromise, without requiring payload inspection or signature-based detection.

Most published work in this space, however, trains and evaluates a detection model within a single collection environment. This leaves open a basic but consequential question: does a behavioral fingerprint learned on one network actually transfer to a different one, with a different device mix, a different topology, and different background traffic? This project set out to answer that question directly, using two independent, publicly available IoT traffic datasets. In the course of doing so, we discovered that the question is harder to answer honestly than it first appears, because the evaluation protocol itself can silently manufacture the appearance of an answer that is not real. This paper reports both the empirical findings and the methodological investigation that corrected them, since we believe the latter is as valuable a contribution as the former.

2. Related Work

Several public datasets have made this kind of research possible. Meidan et al. introduced N-BaIoT, capturing real IoT devices before and during Mirai and BASHLITE infections, and showed that autoencoder reconstruction error can flag the resulting anomalies from network behavior alone [1]. Koroniotis et al. built Bot-IoT to provide more realistic and diverse attack traffic with precise, time-synchronized labels, addressing a limitation of earlier datasets [2]. Garcia et al. released IoT-23, twenty-three labeled captures of real IoT hardware with well-documented labeling methodology, which we use directly in this work [3]. Neto et al. published CICIoT2023, drawing traffic from over one hundred real devices under more than thirty attack types, and this is our second dataset [4]; critically, their own methodology section documents that packet-count features in the public CSV export are computed over a fixed aggregation window (10 or 100 packets) rather than representing true flow length, a detail that turned out to be central to our findings (Section 5.5).

On the modeling side, Yousefnezhad et al. surveyed the broader device fingerprinting space and identified cross-dataset and cross-environment generalization as one of the least examined aspects of the field, since most published results are reported within a single collection environment [5]. NIST IR 8349 provides a formal methodology for characterizing the expected network behavior of IoT devices across their lifecycle [6].

More broadly, the phenomenon we investigate, a model's apparent performance changing when training and test data are drawn from different distributions, is a well-studied problem outside the IoT security literature specifically. Moreno-Torres et al. provide a unifying taxonomy of dataset shift in classification, distinguishing covariate shift, prior probability shift, and concept shift as distinct mechanisms by which train/test mismatch can degrade or distort model performance [9]. Pan and Yang's survey of transfer learning similarly formalizes the assumption, violated by construction in our cross-dataset setting, that training and test data share a common feature space and distribution, and surveys the broader family of techniques developed specifically to address this gap [10]. We situate our findings within this literature: rather than presenting evaluation-protocol-induced artifacts as a novel discovery, we treat them as a concrete, IoT-specific instance of a well-known and actively studied class of problems, and our root-cause finding, that the two datasets' flow-construction methodologies are not just differently scaled but differently defined, is best understood as an especially severe form of covariate shift in the terminology of [9].

3. Methodology

We extract a shared feature schema from each dataset: flow duration, total bytes, total packets, average packet size, and protocol type (TCP, UDP, ICMP, represented as separate binary flags). Because IoT-23 and CICIoT2023 store traffic in entirely different native formats, we built a feature extraction pipeline that reads each dataset's native format and maps it into this common schema.

Three models were implemented. StatBaseline is a per-feature statistical z-score detector: it computes the mean and standard deviation of each continuous feature from benign training traffic, then flags any test flow whose z-score on any single feature exceeds a threshold (3.0 throughout). ClusterBaseline fits K-Means on benign training traffic and flags any flow whose distance to its nearest cluster centroid exceeds the 90th percentile of benign training distances [8]. Isolation Forest isolates anomalies via random recursive partitioning [7]. All three are trained exclusively on benign traffic.

Evaluation proceeds in two modes. Within-environment evaluation holds out 30% of a dataset's benign traffic plus all of its attack traffic as a test set; these results are unaffected by anything discussed in Section 5 and remain valid throughout. Cross-environment evaluation trains on one dataset's benign traffic in full and evaluates on the other. In both modes, every test set is capped to a 1:1 ratio of attack

to benign traffic before scoring, to prevent class imbalance (Section 4.2) from inflating metrics. As detailed in Section 5, our cross-environment standardization approach evolved substantially over the course of this project after we discovered a serious flaw in our initial protocol; the final, corrected protocol standardizes test data using only the training environment's statistics, with no test-side recalibration.

4. Early Challenges and Data-Quality Fixes

Several real obstacles arose while building this pipeline against actual, heterogeneous data. We report them because resolving them correctly, rather than simply obtaining a plausible-looking number, is itself part of this project's contribution.

4.1 Schema inconsistencies in real dataset files

IoT-23's raw conn.log.labeled files use an inconsistent mix of spaces and tabs in the #fields header row, resolved with regular-expression-based whitespace splitting. The merged CICIoT2023 CSV export omitted the documented flow_duration column entirely and capitalized the Label column; we used inter-arrival time (IAT) as a proxy for duration, a substitution later shown to be more than cosmetic (Section 5.5).

4.2 Class imbalance inflating evaluation metrics

Initial cross-environment results showed precision and recall near 1.0 in every scenario. The CICIoT2023 merged files are approximately 97.7% attack traffic (5,578,878 attack rows versus 133,392 benign rows); a model could achieve near-perfect recall simply by flagging nearly every row as an attack. Capping the attack-to-benign ratio at 1:1 in every test set corrected this.

4.3 Feature-scale mismatch across datasets

After correcting class imbalance, cross-environment performance collapsed entirely in one direction (FPR = 1.000). CICIoT2023's IAT values are on the order of tens of millions, while IoT-23's true duration values are single-digit seconds; a z-score threshold learned on one scale is meaningless on another. Our fix at the time was to standardize each environment's features against its own benign reference statistics.

4.4 Reference-set contamination during standardization

An initial attempt computed normalization statistics from a test set containing a mix of benign and attack traffic, letting attacks skew the reference point used to detect them. We corrected this so normalization statistics are computed strictly from benign-only traffic, with cross-environment runs using the target environment's own benign reference.

4.5 Categorical feature instability

Even after the fixes above, one cross-environment direction still caused StatBaseline to flag all traffic as anomalous. CICIoT2023's benign traffic contains zero ICMP packets, producing a standard deviation of zero on the is_icmp feature; any IoT-23 traffic containing ICMP produced an effectively infinite z-score on that feature alone. We resolved this by restricting StatBaseline's z-score calculation to continuous features only.

5. A Deeper Methodological Investigation

With the fixes in Section 4 applied, our pipeline produced the results in Table 3 below: StatBaseline appeared to transfer genuinely well in one direction (F1 = 0.974) while Isolation Forest and ClusterBaseline did not. This section documents why that apparent finding was substantially wrong, how we discovered this, and what the corrected, honest result actually is.

5.1 Within-environment results (valid, unaffected by the investigation below)

Scenario	Model	Precision	Recall / F1 / FPR
Within: IoT-23	StatBaseline	0.249	R=0.009 F1=0.017 FPR=0.159
	IsolationForest	0.796	R=0.385 F1=0.519 FPR=0.099
	ClusterBaseline	0.859	R=0.385 F1=0.532 FPR=0.063
Within: CICIoT2023	StatBaseline	0.972	R=0.976 F1=0.974 FPR=0.030
	IsolationForest	0.693	R=0.226 F1=0.341 FPR=0.100
	ClusterBaseline	0.907	R=0.976 F1=0.940 FPR=0.100

5.2 Initial cross-environment results

Table 3 shows our pipeline's cross-environment output after the Section 4 fixes. Figure 1 shows this result as originally charted.

Configuration	Direction	F1	FPR
StatBaseline (self-calibrated)	IoT-23 → CICIoT2023	0.974	0.031
StatBaseline (self-calibrated)	CICIoT2023 → IoT-23	0.017	—
Isolation Forest (self-calibrated)	IoT-23 → CICIoT2023	0.611	0.999
ClusterBaseline (self-calibrated)	IoT-23 → CICIoT2023	0.667	1.000

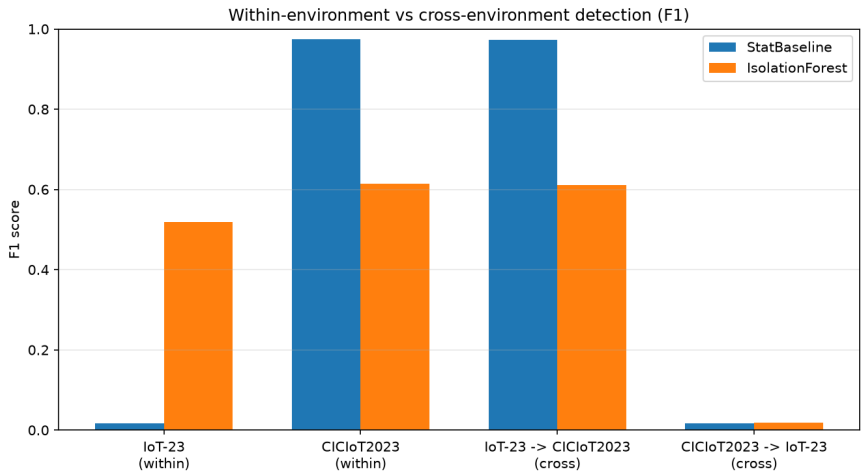


Figure 1. Original cross-environment comparison, later shown in Section 5.3 to substantially misrepresent StatBaseline's result.

5.3 Diagnosing an evaluation artifact

A result this clean, combined with a suspicious numerical pattern (within-environment and cross-environment F1 for the same test dataset were nearly identical regardless of training source: 0.017 ≈ 0.017 for IoT-23, 0.974 ≈ 0.974 for CICIoT2023), prompted us to check whether StatBaseline's trained model actually depended on its training data at all. We verified directly: because our cross-environment protocol standardized training data against its own mean and standard deviation before fitting, StatBaseline's learned mean was always approximately 0 and its learned standard deviation was always exactly 1.0, regardless of which dataset trained it. This is a mathematical certainty, not a coincidence: standardizing any dataset against its own mean and standard deviation always yields

mean 0 and standard deviation 1. StatBaseline's apparent cross-environment transfer was therefore not measuring transfer at all; it was measuring whether the test environment's own attacks are separable from its own benign traffic in self-standardized space, entirely independent of what supposedly trained the model.

5.4 Honest re-evaluation

We corrected the protocol to standardize using only training-derived statistics, applied identically to training and test data, with no test-side recalibration. We additionally tested progressively narrower feature subsets, motivated by the data-quality findings in Section 5.5. Table 4 reports the results.

Feature set (honest, train-only standardization)	IoT-23 → CICIoT2023	CICIoT2023 → IoT-23
All 4 continuous features	F1=0.667, FPR=1.000	F1=0.667, FPR=1.000
Excluding duration	F1=0.667, FPR=1.000	F1=0.666, FPR=1.000
Excluding duration + total_pkts	F1=0.667, FPR=1.000	F1=0.000, FPR=0.000
total_bytes alone	F1=0.000, FPR=0.000	F1=0.000, FPR=0.000

Every configuration collapses to the same degenerate signature: precision near 0.5 and recall near 1.0 (or, in two cases, 0.0 and 0.0) combined with FPR at 1.0 or 0.0, which is the mathematical signature of a model flagging either all or none of a balanced test set rather than discriminating meaningfully. No feature subset, including the two features (total_bytes, avg_pkt_size) that survived every other diagnostic in this paper, produces genuine cross-environment transfer. We verified this finding at the finest available granularity by re-running our per-attack-type breakdown (Section 5.6) under the same honest protocol: every one of 35 individual attack types across both directions showed recall of exactly 1.0, with zero exceptions, confirming that the collapse is total and uniform rather than an average obscuring partial success on some attack types.

5.5 Root cause: incompatible flow-construction methodology

We investigated why even the two most “neutral” features fail to transfer. A feature-distribution comparison revealed that IoT-23's continuous features exhibit extreme kurtosis (e.g., 218,957 for duration, versus 10.37 for the same feature in CICIoT2023) even after standardization. Trimming analysis showed this is not a data-quality defect: the most extreme values (connections lasting up to 22 hours, transferring up to 336MB) are internally consistent, plausible real traffic, not corrupted rows. This reflects the well-known heavy-tailed nature of real, captured network traffic, which a synthetically generated or heavily curated testbed capture may not reproduce to the same degree.

A second, independent problem was more severe. CICIoT2023's benign total_pkts feature has a standard deviation of 0.125, with 99.95% of values exactly equal to 10. This is not a property of real traffic; it is a direct artifact of CICIoT2023's own published feature-extraction methodology, which explicitly groups packets into fixed windows of 10 or 100 for statistical aggregation, rather than measuring actual flow length [4]. IoT-23's total_pkts, by contrast, reflects true per-connection packet counts (93.64% of benign connections have exactly 1 packet, with a legitimate tail reaching 363,141). These are not the same measurement at different scales; they are different measurements of different underlying constructs, and no standardization approach, however sophisticated, can make them comparable without discarding one of them entirely.

5.6 Per-attack-type confirmation

Under the original (flawed) self-calibrated protocol, our per-attack breakdown showed a striking split: all 19 flooding attack types (DDoS, DoS, Mirai variants) achieved StatBaseline recall of 1.000, while all 13 reconnaissance and injection attack types achieved recall between 0.023 and 0.304. Figure 2 shows a representative subset of this split, spanning the full observed range.

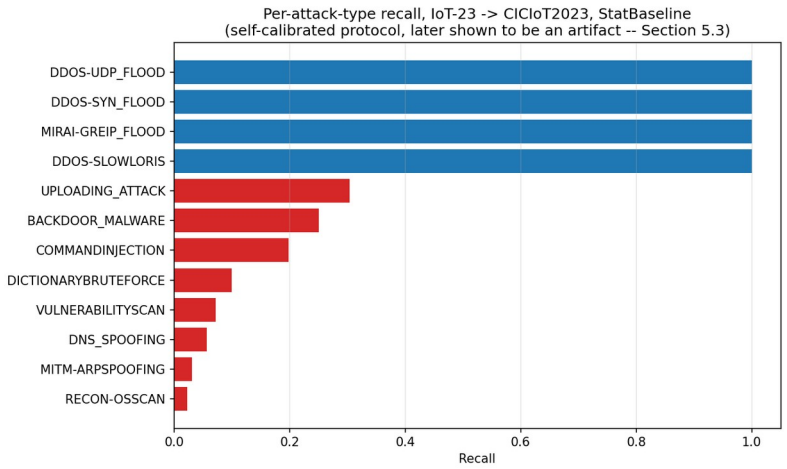


Figure 2. Representative per-attack-type recall under the original self-calibrated protocol. This pattern is explained as an artifact below, not a genuine cross-environment finding.

Under the corrected honest protocol, this distinction entirely disappears: all 35 attack types, floods and reconnaissance alike, show recall of exactly 1.0 in both directions. We conclude that the flood-versus-reconnaissance asymmetry visible in Figure 2 was itself a byproduct of the self-calibration artifact, specifically equivalent to a within-CICIoT2023 (respectively within-IoT-23) breakdown of attack separability, not evidence of differential cross-environment transfer by attack type.

5.7 Does this artifact affect Isolation Forest and ClusterBaseline equally?

Because Isolation Forest and ClusterBaseline are not reducible to two numbers the way StatBaseline is, we hypothesized they would be immune to the degeneracy in Section 5.3. We tested this directly: for each model, we fit two instances, one on each dataset's self-standardized benign traffic, and applied both to an identical, fixed set of test points, measuring how often the two instances agree. Near-total agreement would indicate the same training-independence problem; substantial disagreement would indicate genuine, training-specific structure. Table 5 reports the result.

Model	Agreement rate*	Interpretation
StatBaseline	~1.000 (by construction)	Fully degenerate
Isolation Forest	0.356	Genuinely differentiated
ClusterBaseline (K-Means)	0.948	Substantially degenerate

**Fraction of identical predictions between a model trained on IoT-23 and an otherwise identical model trained on CICIoT2023, evaluated on the same fixed test points.*

Isolation Forest shows 64.4% disagreement between the two differently-trained instances, confirming its cross-environment failure (Table 3) reflects genuine, verified structural mismatch between training and test geometry, not an evaluation artifact. ClusterBaseline, however, shows 94.8% agreement, indicating its result is substantially, though not completely, subject to the same self-calibration problem that fully explained StatBaseline's behavior. We attribute this to K-Means with a small, fixed number of clusters producing broadly similar relative partition geometry on any standardized low-dimensional data, regardless of the data's true underlying shape, unlike Isolation Forest's splits, which are more sensitive to the specific value distribution used to build them. We report this honestly as a partial confound rather than treating ClusterBaseline's collapse as equally diagnostic as Isolation Forest's.

5.8 A complementary result: within-dataset scenario fingerprinting

The investigation above addresses cross-dataset transfer specifically. A related but distinct question from our original project scope, whether behavioral features can distinguish different device or capture sources at all, remains answerable entirely within one dataset, sidestepping the incompatibility

documented above. We trained a Random Forest classifier to predict which of IoT-23's five scenario source files a benign flow originated from, using the same feature set as the rest of this paper. We label this “scenario classification” rather than “device model fingerprinting,” since we have not separately verified the specific real-world device behind each scenario file against IoT-23's own documentation.

Overall accuracy was 0.994, but this is misleading on its own: one scenario (malware1) contains 98.2% of all benign rows, so a classifier that always guesses that scenario would already score 0.982. The appropriate metric is the macro-averaged F1 across all five scenarios, treating each equally regardless of size, which was 0.88 against a random-chance baseline of 0.20. Table 6 reports per-scenario results.

Scenario	Precision	Recall	F1
malware1	1.00	1.00	1.00
honeypot4	0.93	0.96	0.95
malware8	0.89	0.87	0.88
honeypot5	0.96	0.73	0.83
malware3	0.85	0.64	0.73
Macro average	0.92	0.84	0.88

Three of five scenarios (malware1, honeypot4, malware8) are strongly distinguishable (F1 ≥ 0.88); two (honeypot5, malware3) are substantially weaker, particularly in recall, indicating these sources' benign traffic is more frequently confused with other scenarios. Feature importance analysis showed connection duration alone accounts for 59.7% of the Random Forest's decision weight, with total_pkts (17.4%), avg_pkt_size (9.4%), and total_bytes (9.0%) contributing modestly, and protocol flags contributing less than 2% combined. We conclude that, within a single consistently-constructed dataset, IoT device or capture scenarios do carry genuine, learnable behavioral fingerprints, and that this signal is carried almost entirely by connection timing rather than protocol choice, a finding that is methodologically independent of, and unaffected by, the cross-dataset incompatibility documented in Sections 5.3–5.5.

Figure 3 summarizes the central corrected finding of this paper: within-environment detection is genuinely effective, while honest cross-environment evaluation collapses to a flagged-everything (or flagged-nothing) degenerate state, visible specifically in the false-positive rate rather than F1 alone.

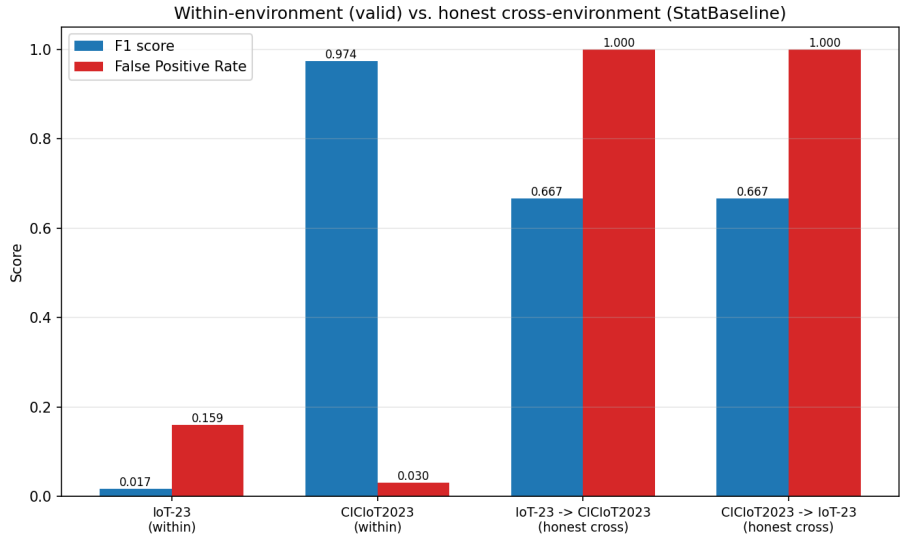


Figure 3. Within-environment results remain valid; honest cross-environment evaluation collapses, visible in FPR even where F1 alone would look moderate.

6. Limitations

The substitution of inter-arrival time for true flow duration in CICIoT2023 (Section 4.1) turned out to be a more serious limitation than initially assessed; Section 5.5 shows the two datasets' packet-count features are not merely differently scaled but differently defined, which likely generalizes to other feature pairs we did not individually audit. Our scenario-fingerprinting result (Section 5.8) is based on five scenarios within a single dataset and does not establish that device fingerprints generalize across datasets, consistent with the paper's central finding. The Isolation Forest and ClusterBaseline hyperparameters (contamination = 0.1, n_clusters = 8) were fixed throughout rather than tuned; given that the collapse we document is driven by feature incompatibility rather than threshold miscalibration, we have no reason to expect tuning would change the qualitative conclusion, but we did not verify this directly. Finally, we did not extend our modeling comparison to autoencoder-based approaches, and our scenario-fingerprinting label is deliberately scoped to “capture source” rather than verified real-world device identity. We view these as defensible scope decisions given the semester timeline, stated explicitly as boundaries on what this work establishes.

7. Conclusion

This project set out to test whether IoT behavioral fingerprints transfer across network environments. Our initial results appeared to answer this question affirmatively for a simple statistical baseline. On investigation, this answer was substantially wrong: it was an artifact of an evaluation protocol that let a simple model's parameters collapse to a dataset-independent fixed point, making its result a measurement of the test environment's own internal structure rather than genuine transfer. Once corrected, we found that none of three tested models, statistical, clustering-based, or tree-based, achieve genuine cross-environment transfer between IoT-23 and CICIoT2023, under any feature subset we tested. We traced this to a root cause more fundamental than feature scaling: the two datasets' packet-count features are not the same measurement, because CICIoT2023's own published methodology aggregates packets into a fixed window rather than counting true flow length. A follow-up diagnostic showed this evaluation artifact itself affects models unevenly, fully explaining the statistical baseline's false success, substantially (though not fully) affecting the clustering model, and not affecting Isolation Forest, whose genuine cross-environment failure we independently verified. A complementary within-dataset experiment confirmed that real, learnable behavioral fingerprints do exist for distinguishing IoT-23's own capture scenarios, carried primarily by connection timing. We report the complete corrected investigation, including the parts where we were initially wrong, as the central contribution of this work: evaluating cross-environment generalization in IoT anomaly detection requires actively verifying that an evaluation protocol is not itself manufacturing the result being measured, a risk the broader dataset-shift and transfer-learning literature has long identified [9], [10] but which is easy to overlook in a from-scratch implementation.

References

[1] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, D. Breitenbacher, A. Shabtai, and Y. Elovici, “N-BaIoT—Network-based detection of IoT botnet attacks using deep autoencoders,” IEEE Pervasive Computing, vol. 17, no. 3, pp. 12–22, 2018.

[2] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, “Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset,” Future Generation Computer Systems, vol. 100, pp. 779–796, 2019.

[3] S. Garcia, A. Parmisano, and M. J. Erquiaga, “IoT-23: A labeled dataset with malicious and benign IoT network traffic,” Stratosphere Laboratory, CTU University, 2020.

[4] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, “CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment,” Sensors, vol. 23, no. 13, p. 5941, 2023.

[5] M. Yousefnezhad, J. Malm, A. Andrae, and others, “A survey on device behavior fingerprinting: Data sources, techniques, application scenarios, and datasets,” arXiv preprint arXiv:2008.03343, 2020.

[6] National Institute of Standards and Technology, NIST IR 8349: Methodology for Characterizing Network Behavior of Internet of Things Devices, Aug. 2025.

[7] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation Forest,” in Proc. 2008 Eighth IEEE Int. Conf. on Data Mining (ICDM), 2008, pp. 413–422.

[8] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” in Proc. 5th Berkeley Symp. on Mathematical Statistics and Probability, vol. 1, 1967, pp. 281–297.

[9] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, “A unifying view on dataset shift in classification,” Pattern Recognition, vol. 45, no. 1, pp. 521–530, 2012.

[10] S. J. Pan and Q. Yang, “A survey on transfer learning,” IEEE Transactions on Knowledge and Data Engineering, vol. 22, no. 10, pp. 1345–1359, 2010.